

ENTHUSIAST ADVANCE COURSE

PHASE : MED-I (B)

TARGET : PRE MEDICAL 2025

Test Type : MINOR

Test Pattern : NEET (UG)

TEST DATE : 11-02-2024

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	4	4	3	1	1	2	3	1	2	2	3	4	3	1	1	4	3	4	2	1	1	2	3	3	4	2	3	2	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	4	2	3	2	3	3	2	4	2	2	1	1	2	1	1	2	4	1	4	2	2	1	2	4	1	2	2	4	4	3
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	2	4	4	2	1	2	3	2	1	2	1	2	2	2	3	2	1	2	3	3	2	1	1	3	3	2	4	3	4
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	3	4	3	3	2	2	4	3	4	2	3	4	4	4	2	2	3	3	1	2	2	2	3	1	1	3	3	3	1
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	3	2	3	1	1	2	2	3	3	1	3	2	3	2	1	1	1	1	4	3	4	1	3	2	2	2	1	2	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	3	4	3	3	2	1	3	1	2	4	3	3	4	3	1	3	2	3	3	2	2	4	4	2	4	2	3	4	4
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	3	1	4	3	2	2	2	1	4	4	3	2	1	3	3	3	2	2	2	2										

HINT - SHEET

SUBJECT : PHYSICS

SECTION-A

1. **Ans (4)**

Gravitational force is same irrespective of medium between masses, whereas electrostatic force depends on medium of charges.

2. **Ans (4)**

A acquires (+ve) and B acquires (-ve) charge.

3. **Ans (4)**

Mass may increase or decrease

4. **Ans (3)**

Here for two sphere $\Delta V=0$

$$\left(\because v_1 = \frac{K(9Q)}{R}, v_2 = \frac{K(9Q)}{R} \right)$$

$\Rightarrow$ No flows of charge takes place

6. **Ans (1)**

Force on both the charges will be equal in magnitude but opposite in direction.

$$F_{12} = F_{21}$$

$$\text{So, } \frac{F_{12}}{F_{21}} = \frac{1}{1}$$

7. **Ans (2)**

$$F_m = \frac{F_{\text{vacuum}}}{K} = \frac{20}{5} = 4N$$

8. **Ans (3)**

$$10 = \frac{K(3)(4)}{r^2} = \frac{12K}{r^2}$$

$$F = k \frac{(3-6)(4-6)}{r^2} = \frac{k(-3)(-2)}{r^2} = \frac{6k}{r^2}$$

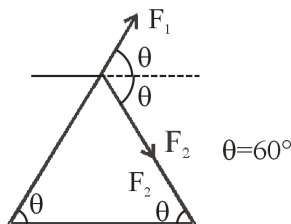
$$\Rightarrow \frac{F}{10} = \frac{6k/r^2}{12k/r^2} = \frac{1}{2} \Rightarrow F = 5N$$

9. **Ans (1)**

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$\epsilon_0 = \frac{q_1 q_2}{F r^2} = \frac{[AT][AT]}{[MLT^{-2}][L^2]} = M^{-1}L^{-3}T^4A^2$$

10. Ans (2)

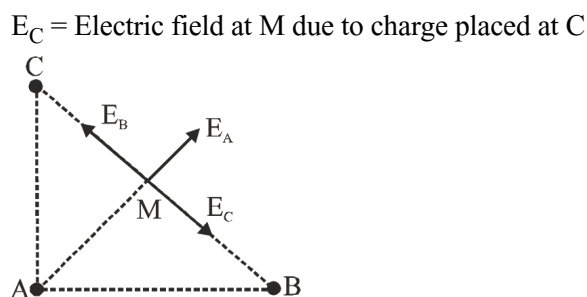


$$F_1 = F_2 = \frac{9 \times 10^9 \times 5 \times 10^{-6} \times 10 \times 10^{-6}}{(0.05)^2} = 180 \text{ N}$$

$$F_{\text{net}} = 2F \cos \theta = 2F \cos 60^\circ = F = 180 \text{ N}$$

11. Ans (2)

E_A = Electric field at M due to charge placed at A
 E_B = Electric field at M due to charge placed at B
 E_C = Electric field at M due to charge placed at C



As seen from figure $|\vec{E}_B| = |\vec{E}_C|$, so net electric field at M, $E_{\text{net}} = E_A$; in the direction of vector 2.

13. Ans (4)

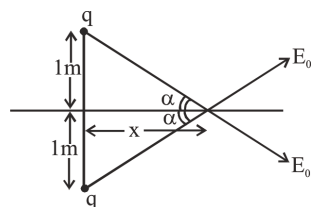
$$\text{Before transfer of charge, } F = \frac{Kq^2}{r^2}$$

After transfer $F' =$

$$\frac{K \left(\frac{q}{2} \right) \left| \left(-q + \frac{q}{2} \right) \right|}{r^2} = \frac{Kq^2}{4r^2} = \frac{F}{4}$$

14. Ans (3)

As two particle A and B having equal charges



$$E_{\text{net}} = 2E_0 \cos \alpha = 2 \times \frac{1}{4\pi\epsilon_0} \frac{q}{\sqrt{1+x^2}} \times \frac{x}{(1+x^2)}$$

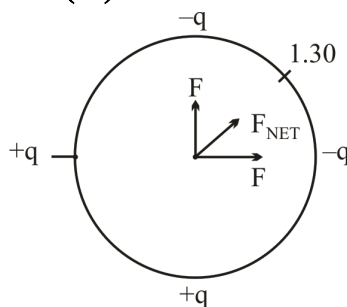
$$= \frac{2qx}{4\pi\epsilon_0(x^2+1)^{3/2}}$$

$$F_{\text{net}} = q_0 E_{\text{net}} = \frac{2qq_0x}{4\pi\epsilon_0(x^2+1)^{3/2}}$$

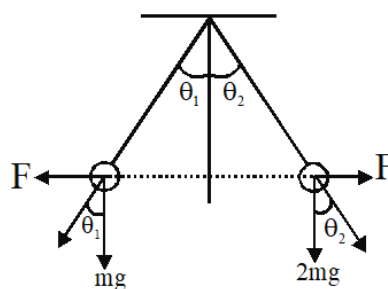
$$\text{at } x = \frac{1}{\sqrt{2}} \quad \frac{dF}{dx} = 0$$

15. Ans (1)

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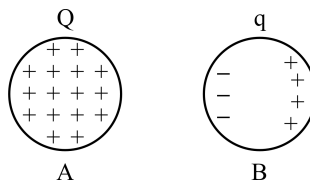
16. Ans (1)



It is clear from figure that θ_1 is greater than θ_2

17. Ans (4)

Two similar charged bodies may attract each other only in special case.



If Q and q are similar charge and induced charge of body B is at smaller distance to A as compare to positive charge of B. If $q < Q$ there is a possibility that attraction is more than repulsion.

19. Ans (4)

Since the incident ray is between focus & pole, reflected ray will be '4'.

20. Ans (2)

$$\text{time} = \frac{tu}{c} = \frac{2 \times 10^{-3} \times 1.5}{3 \times 10^8}$$

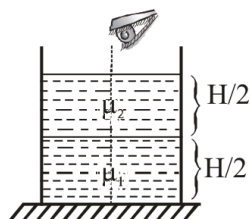
21. Ans (1)

$$\mu = \frac{h_1}{h_2} = \frac{1.5}{2} = \frac{5}{4}$$

$$\frac{h_2}{h_1} = \frac{x + y/\mu}{y + \mu x}$$

22. Ans (1)

$$\text{Apparent Depth} = \frac{H/2}{\mu_1} + \frac{H/2}{\mu_2}$$



23. Ans (2)

If object is moved away from mirror i.e. from its position then size decreases become image always form between pole and focus.

24. Ans (3)

In case of convex mirror when object moves towards the mirror then image approaches from infinity to pole.

25. Ans (3)

$$\text{Angle} = \frac{\text{Arc}}{\text{Radius}}$$

$$0.5 \times \frac{\pi}{180} = \frac{\text{Arc}}{7.5 \times 100}$$

26. Ans (4)

$$\frac{1}{f} = \frac{1}{-15} + \frac{1}{6} = \frac{-2+5}{30}$$

$$f = 10 \text{ cm}$$

27. Ans (2)

$$\mu = \frac{c}{v} \Rightarrow 1.5 = \frac{3 \times 10^8}{v}$$

28. Ans (3)

$$\sin \theta_c = \frac{1}{R\mu_D}; \left[\because \mu \propto \frac{1}{\lambda} \right]$$

$$\sin \theta_c = \frac{\lambda_D}{\lambda_R} = \frac{3500\text{\AA}}{7000\text{\AA}}; \sin \theta_c = \frac{1}{2}; \theta_c = 30^\circ$$

29. Ans (2)

$$\text{Shifting} = t \left(1 - \frac{1}{\mu} \right) = 12 \left(1 - \frac{1}{1.5} \right)$$

$$\text{Shifting} = 4 \text{ cm}$$

30. Ans (3)

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$$v = \lambda \times \nu \text{ and } v = \frac{c}{\mu}$$

$$5000 \times 10^{-10} \times 4 \times 10^{14} = \frac{3 \times 10^8}{\mu}$$

$$\Rightarrow \mu = 1.5$$

32. Ans (2)

$$LC = \frac{\text{Pitch}}{\text{Number of CSD}}$$

$$\text{Pitch} = LC \times \text{Number of CSD}$$

$$= (0.02 \text{ mm}) \times 100 = 2 \text{ mm}$$

33. Ans (3)

$$\therefore R = \frac{V}{I} = \frac{100}{20} = 5 \Omega$$

$$\text{also } \frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I} = \frac{5}{100} + \frac{0.4}{20} = \frac{7}{100}$$

$$\Delta R = \frac{7}{100} \times R = \frac{7}{20} = 0.35$$

rounding off according to significant figures $\Delta R = 0.4$.

$$\text{so } R = (5 \pm 0.4) \Omega$$

34. Ans (2)

$$LC = MSD - VSD$$

$$LC = 0.1 \text{ mm}$$

Zero error = main scale reading + (LC) (ver. scale reading)

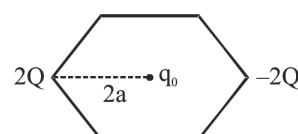
$$= 0 + (0.1 \text{ mm}) 6$$

$$= 0.6 \text{ mm}$$

SECTION-B

36. Ans (3)

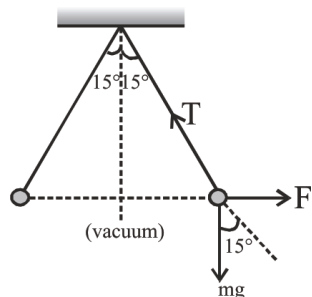
By using reduction method, we can write



so, Force on q_0 is

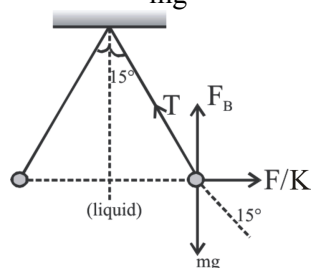
$$F = \frac{k(2Q)q_0}{(2a)^2} = \frac{KQq_0}{2a^2}$$

37. Ans (2)



The ball is in equilibrium under three forces as shown.

$$\tan 15^\circ = \frac{F}{mg} \quad \dots\dots(i)$$



In liquid, electric force is F/K ,

where K : dielectric constant of liquid.

Upthrust F_B acting on ball in upward direction

$$F_B = \rho V g = \rho \frac{m}{\sigma} g$$

where ρ ; density of liquid,

σ ; density of material

$$\begin{aligned} \tan 15^\circ &= \frac{F/K}{mg - F_B} = \frac{F/K}{mg - \frac{\rho mg}{\sigma}} \\ &= \frac{F/K}{mg \left(1 - \frac{\rho}{\sigma}\right)} \quad \dots\dots(ii) \end{aligned}$$

From (i) and (ii)

$$K = \frac{1}{1 - \frac{\rho}{\sigma}} = \frac{1}{1 - \frac{0.8}{1.6}} = 2$$

38. Ans (4)

$$E = \frac{kq}{r^2} \text{ where}$$

$$\begin{aligned} r &= |\vec{r}_1 - \vec{r}_0| = |4\hat{i} + 5\hat{j} - \hat{i} - \hat{j}| \\ &= |3\hat{i} + 4\hat{j}| = 5m \end{aligned}$$

$$\Rightarrow E = \frac{9 \times 10^9 \times 25 \times 10^{-6}}{5 \times 5} = 9 \times 10^3 \text{ V m}^{-1} = \frac{9 \text{ kV}}{m}$$

39. Ans (2)

By theory

40. Ans (2)

Apply superposition of electric field using the fact that electric field is radially outwards.

41. Ans (1)

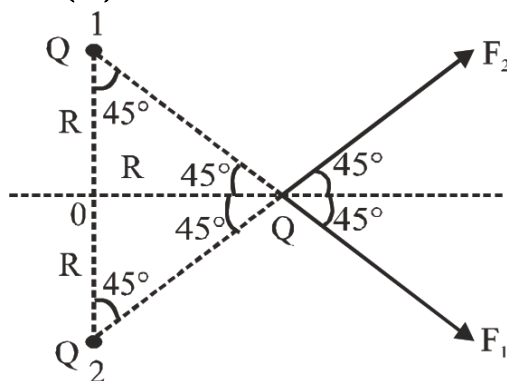
$$Q_1 = 2\sqrt{2}\mu\text{C}; \vec{r}_1 = (\hat{i} + 2\hat{j} + 3\hat{k})m$$

$$Q_2 = 5\mu\text{C}; \vec{r}_2 = (4\hat{i} + 6\hat{j} + 8\hat{k})m$$

$$\vec{r}_{21} = 3\hat{i} + 4\hat{j} + 5\hat{k}; |\vec{r}_{21}| = \sqrt{50}m$$

$$\begin{aligned} \vec{F}_{21} &= \frac{kQ_1Q_2}{|\vec{r}_{21}|^2} \hat{r}_{21} = \frac{kQ_1Q_2}{|\vec{r}_{21}|^3} \vec{r}_{21} \\ \vec{F}_{21} &= \frac{9 \times 10^9 \times 2\sqrt{2} \times 10^{-6} \times 5 \times 10^{-6}}{(\sqrt{50})^3} (3\hat{i} + 4\hat{j} + 5\hat{k}) \\ &= 36 \times 10^{-5} (3\hat{i} + 4\hat{j} + 5\hat{k})\text{N} \end{aligned}$$

42. Ans (1)



$$F_{\text{Net}} = \sqrt{F_1^2 + F_2^2}$$

$$F_1 = F_2 = \frac{kQ \cdot Q}{(R\sqrt{2})^2} = \frac{kQ^2}{2R^2}$$

$$F_{\text{Net}} = \sqrt{2} \frac{kQ^2}{2R^2} = \frac{\sqrt{2}Q^2}{8\pi\epsilon_0 R^2}$$

43. Ans (2)

$$A = 4d^2$$

$$\pi \left[\frac{d}{\sqrt{u^2 - 1}} \right]^2 = 4d^2$$

$$\frac{\pi}{u^2 - 1} = 4$$

$$u = \frac{\sqrt{\pi + 4}}{2}$$

44. Ans (1)

$$\begin{aligned}\text{Shift} &= t \left(1 - \frac{1}{r\mu_d} \right) \\ &= t \left(1 - \frac{V_d}{V_r} \right) \\ &= 18 \left(1 - \frac{1.5 \times 10^8}{\frac{9}{4} \times 10^8} \right) \\ &= 6 \text{ cm}\end{aligned}$$

45. Ans (1)

$$\begin{aligned}m &= \frac{f}{f-u} \Rightarrow 3 = \frac{f}{f+2} \\ f &= -3 \text{ cm} \\ f &= \text{Concave mirror, } -3 \text{ cm}\end{aligned}$$

46. Ans (2)

$$\begin{aligned}V_I &= m^2 V_0 \\ &= \left[\frac{10}{10+50} \right]^2 \times 16 \\ &= \frac{1}{36} \times 16 \\ &= \frac{4}{9} \text{ m/s}\end{aligned}$$

47. Ans (4)

$$\begin{aligned}m &= \frac{f}{f-u} \Rightarrow m_1 = \frac{f}{f+25} \quad m_2 = \frac{f}{f+40} \\ \text{on dividing} \quad \frac{m_1}{m_2} &= \frac{f+40}{f+25} = 4 \Rightarrow f = -20 \text{ cm}\end{aligned}$$

48. Ans (1)

$$\begin{aligned}m &= -2 = \frac{f}{f-u} \\ 2f - 2u &= -f \\ 3f &= 24 \\ u &= \frac{3f}{2}\end{aligned}$$

50. Ans (2)

$$\begin{aligned}\text{L.C.} &= M \left[1 - \frac{8}{10} \right] \\ &= \frac{1}{10} \left[\frac{2}{10} \right] \text{ cm} = 0.02 \text{ cm}\end{aligned}$$

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SUBJECT : CHEMISTRY

SECTION-A

54. Ans (4)

$$\begin{aligned}\text{Given 0 order, } K &= 2 \times 10^{-2} \frac{\text{mol}}{\text{L sec}} \\ t &= 25 \text{ sec, } (A)_t = 0.5 \text{ M, } [A]_0 = ? \\ [A]_t &= [A]_0 - Kt \\ 0.5 &= [A]_0 - 2 \times 10^{-2} \times 25 \\ [A]_0 &= 1 \text{ M}\end{aligned}$$

59. Ans (4)

$$\begin{aligned}\text{NCERT Pg. \# 98} \\ \text{ROR} &= -\frac{d[A]}{dt} = -\frac{[0.3 - 0.8]}{5 \times 2} = \frac{0.5}{5 \times 2} \\ &= \frac{0.1}{2} = 5 \times 10^{-2} \text{ M. min}^{-1}\end{aligned}$$

60. Ans (3)

NCERT, Pg. \# 98

63. Ans (4)

$$\begin{aligned}\text{ROR}(r) = K &= \frac{(\text{ROA})_{H_2}}{3} \\ (\text{ROA})_{H_2} &= 3 \times 5 \times 10^{-4} \\ &= 15 \times 10^{-4} \text{ mol, L}^{-1} \text{ s}^{-1}\end{aligned}$$

65. Ans (2)

Rate of reaction does not depend on concentration of excess reactant.

67. Ans (2)

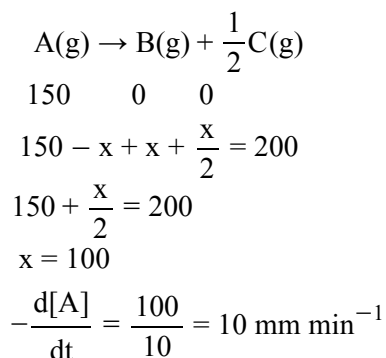
$$\text{Order} = \frac{1}{2} - 1 + \frac{3}{2} = 1$$

SECTION-B

90. Ans (4)

$$\begin{aligned}\log \left(\frac{K}{A} \right) &= \frac{E_a}{2.303RT} \\ \log(10^{-5}) &= \frac{E_a}{2.303 \times 2 \times 300} \\ -5 &= \frac{-E_a}{2.303 \times 2 \times 300} \\ E_a &= 2.303 \times 2 \times 300 \times 5 \\ &= 6909 \text{ cal} = 6.909 \text{ Kcal}\end{aligned}$$

91. Ans (2)

**SUBJECT : BIOLOGY-I****SECTION-A**

101. Ans (2)

NCERT-XII, Pg. # 96

102. Ans (3)

NCERT (XI) Pg. # 144

103. Ans (4)

NCERT (XI) Pg. # 145, Fig. 9.1

104. Ans (4)

NCERT (XII) Pg. # 48, para 2

105. Ans (4)

NCERT (XIIth) Pg. # 152 Para -1

106. Ans (2)

NCERT (XII) Pg. # 97,98,100

107. Ans (2)

NCERT XII, Pg. # 107

108. Ans (3)

NCERT-XII, Pg # 102

109. Ans (3)

NCERT Pg. # 106, Fig. 6.4.2

110. Ans (1)

NCERT XII Pg, 103

111. Ans (2)

NCERT XII, Pg. # 106, 6.4.2

112. Ans (2)

NCERT XII Pg. 99

113. Ans (2)

NCERT XII Pg. # 99

114. Ans (3)

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NCERT (XII) Pg. # 99(E), 108(H)

115. Ans (1)

NCERT Pg. # 144, 145

116. Ans (1)

NCERT XII Pg # 102

117. Ans (3)

NCERT Pg # 96

118. Ans (3)

NCERT Pg. # 105

119. Ans (3)

NCERT Pg. # 96

120. Ans (1)

NCERT (XIIth) Pg. # 108

121. Ans (2)

NCERT-XII Pg. # 111

122. Ans (3)

NCERT, Pg#151, 2nd Last Para.

123. Ans (2)

NCERT Pg # 144, last paragraph

124. Ans (3)

NCERT XII Page No. # 97

125. Ans (1)

NCERT Pg. # 107(E), Fig. 6.8
Pg. # 115(H), Fig-6.8

126. Ans (1)

NCERT Pg. # 152

127. Ans (2)

NCERT Pg. # 152

128. Ans (2)

NCERT Pg. # 152

129. Ans (3)

NCERT Pg. # 151

130. Ans (3)

NCERT (XII) Pg. # 103

131. Ans (1)

Module-09, Pg. No.-111

132. Ans (3)

NCERT, Pg. # 100-Search for Genetic Material

133. **Ans (2)**
NCERT Pg # 103

134. **Ans (3)**
NCERT XII Pg. 105

135. **Ans (2)**
NCERT Page-106

SECTION-B

136. **Ans (1)**
NCERT Pg. # 144 last para

137. **Ans (1)**
NCERT (XI) Pg. # 152

138. **Ans (1)**
NCERT XII, pg.# 101

139. **Ans (1)**
NCERT (XII) Pg # 107, Fig. 6.8

140. **Ans (4)**
NCERT Pg # 103

141. **Ans (3)**
NCERT (XII) Pg. # 100

142. **Ans (4)**
NCERT XII Pg. # 96, para-6.1.1

143. **Ans (1)**
NCERT Pg. # 151

144. **Ans (3)**
NCERT, Pg. # 103

145. **Ans (2)**
NCERT (XI) Pg. # 149,151,152

146. **Ans (2)**
NCERT XII, Pg # 119-120

147. **Ans (2)**
NCERT-XI, Pg # 145, Fig. 9.1

148. **Ans (1)**
NCERT Pg. # 96

149. **Ans (2)**
NCERT (XIIth) Pg. # 97

150. **Ans (2)**
NCERT XII Pg. # 105

SUBJECT : BIOLOGY-II

SECTION-A

159. **Ans (1)**
NCERT Pg. # 51

160. **Ans (2)**
NCERT XII, Pg # 43, para-3.1

162. **Ans (3)**
NCERT-XII Pg. # 48

163. **Ans (3)**
NCERT Pg. No. # 48

164. **Ans (4)**
Colostrum is nutritious contains proteins, several antibody, essential fatty acids for proper development of baby.

166. **Ans (1)**
NCERT: Pg #: 46

169. **Ans (3)**
NCERT Pg. No. 30

175. **Ans (2)**
NCERT-XII Page#47

179. **Ans (4)**
NCERT Pg. # 50

180. **Ans (4)**
NCERT-XII, Pg. # 50, Fig. 3.9 (E)
NCERT-XII, Pg. # 54, Fig. 3.9 (H)

184. **Ans (3)**
NCERT (XIIth) Pg. # 50

185. **Ans (2)**
NCERT(XII) Pg#51/55(H) Para:3.5

SECTION-B

191. **Ans (3)**
Middle part of sperm possesses numerous mitochondria for energy.

192. **Ans (2)**
Both statements are correct.

195. **Ans (3)**
NCERT-XII, Pg. # 49

199. **Ans (2)**
NCERT (XIIth) (E)Pg. # 48, fig-3.6